

The Compu-Global-Hyper-Mega-Net as a Discourse Community

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A group of people can make a community. They might share similar interests, or they may live in the same town. But beyond that simplistic idea of community, there is something called a “discourse community.” There are several criteria for a simple group of people to become a discourse community. These criteria, as defined by John Swales, and paraphrased by Mezler (2020), are:

1. A broadly agreed upon set of common public goals
2. Mechanisms of intercommunication among members
3. Use of these communication mechanisms to provide information and feedback
4. One or more genres that help further the goals of the discourse community
5. A specific Lexis (specialized language)
6. A threshold level of expert members (24-26) (p. 102).

These six criteria ensure that the members of the community are active participants; they collaborate and communicate with one another (they have discourse). These communities need to be able to grow, to learn off of each other, and to expand their niches beyond just an insular group. I of course am a part of a discourse community. I could just say that discourse community is “retro computing,” as in people who collect and mess with obsolete computers, but that’s too vague. There are just too many different types of retro computing enthusiasts. Take for instance, users on the online forum Reddit, who make up the “subreddit” r/amiga. The group very clearly has a common set of public goals, as stated by the subreddit’s description: “We talk about relevant news, events, new and old hardware and software and all things Commodore Amiga” (r/amiga, 2008). Even though a user in r/amiga and me are into old computers, we aren’t in the

same discourse community, as we aren't interested into the same type of old computers and are unlikely to interact because of it.

What is my discourse community?

I've been spending a lot of time recently interacting with the people who take part in a project called the "Compu-Global-Hyper-Mega-Net" (CGHMN). This project is an effort to create a private "retro internet" for old computers and operating systems. These obsolete systems have massive security flaws and general functionality issues when used in modern networks and the modern internet. So the CGHMN was created to avoid these issues while also allowing old systems to talk to each other over the internet in a safe manner. To explore how well CGHMN operates as a discourse community as defined by Swales and Melzer, I'll start with the shared common goals of the community. Then I'll explore the different methods of communication that the other members of the community use, and how those communication methods are used for information and feedback. And building on that last step, the next part is exploring the different forms the community takes during communication and presentation, and how our specialized languages are integral to the discourse community. The penultimate step is investigating the relationship between novice and expert members; how could a new person learn, and what types of members are valued? And finally, I'll explore how I interact with the CGHMN and some other aspects of discourse communities.

A broadly agreed upon set of public goals

This aspect of Swales' definition is probably the most important, as expressed by my r/amiga subreddit example. A broad shared interest, like retro computing, can be shared by many different people, but if a group of those people does not have a shared goal, there isn't a reason for them to interact, so they stay isolated and do not collaborate. Or as Melzer (2020) explains

when comparing guitar players, people who like music, and guitarists in Sacramento California:

“They don’t share the same goals, and they don’t all interact with each other to meet the same goals” (p. 102). A community needs to have a reason to interact and hold discourse to really be a discourse community. The CGHMN’s shared goal can be found on the project’s personal wiki page, <https://wiki.cursedsilicon.net/wiki/Compu-Global-Hyper-Mega-Net> (Figure 1).

Figure 1

<https://wiki.cursedsilicon.net/wiki/Compu-Global-Hyper-Mega-Net>

Preamble

Compu-Global-Hyper-Mega-Net (hereby shortened to **CGHMN** for easier reference) originally started as me making good on a promise to use a Cisco AS5300 I acquired years ago to make a Dial-Up internet provider. Over time while working on the project and watching other YouTubers work on their own ISP projects it became clear that many of them are happy to set up the hardware, there was very little interest (or ability) to make it usable to other folks, or to have anything fun or useful to "do" with it (beyond the novelty of browsing a few vintage websites such as FrogFinder)

CGHMN (its name [borrowing from a Simpsons joke, naturally](#)) is another one of my "biting off far more than I can possibly chew" projects. Intended to be an intersection point between art, shitpost, rejection of "internet modernity", refuge for the nostalgic and a communal home for projects of a specifically retro computing bent.

CGHMN is currently in [Open Beta](#). Users can sign up and join the network along with their devices. While the network is totally solid, we need a large enough userbase to make it interesting for users to contribute! A bit of a chicken-and-egg problem. But hopefully one we can overcome.

An eventual goal is to provide "open" access to an era of the web that has been not so much "lost" as *obliterated* by capitalism. A space that can mimic "the vibe" of the late 1990's internet through faithful reconstruction or mimicry of the hardware and software stacks that powered it.

Mechanisms will be implemented to discourage (but not outright prevent) "modern systems" from accessing the network where possible, both for the security of users on the network and to try and further the "vibe" of using period-accurate hardware and software. Using IRC and playing StarCraft on Windows 11 simply isn't as *fun* for us!

So what's this all about?

CGHMN is a collaborative project aimed to create a late 1990's/early 2000's compatible "internet" for retro computing enthusiasts and their machines to connect to each other

Note. This is a screenshot of the article for the CGHMN on wiki.cursedssilicon.net, with only the sections for the preamble and “So what’s this all about?” included.

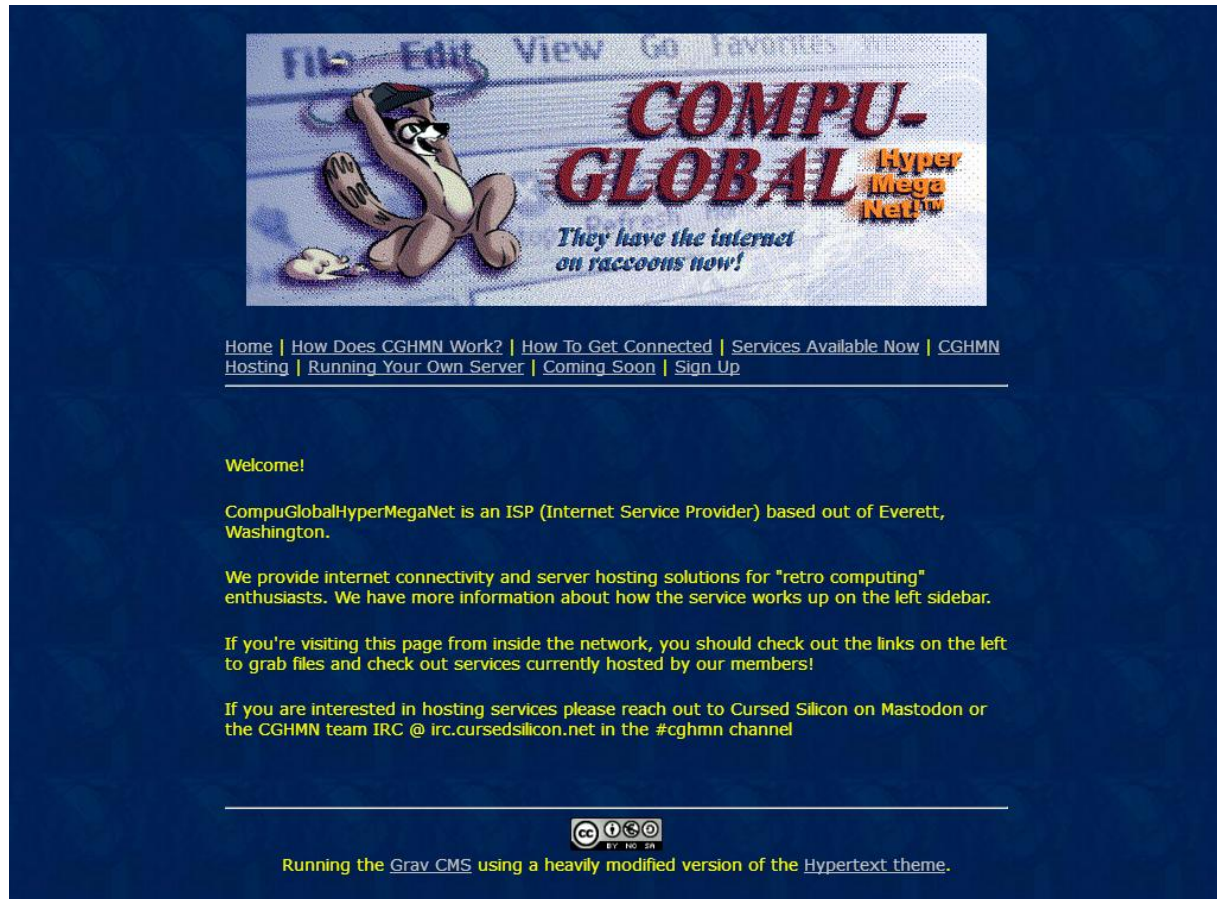
The stated goal of the CGHMN is to “create a late 1990’s/early 2000’s compatible ‘internet’ for retro computing enthusiasts and their machines to connect to each other” (Figure 1). But the page also says that some other goals are to expand the userbase, only have era-appropriate systems and services running on the network, and to share complicated retro networking concepts (such as dial-up internet access) to as many people as possible. These goals give retro computing enthusiasts who happen upon the project something to work for, play with, and collaborate on.

Mechanisms of intercommunication among members and use of these communication mechanisms to provide information and feedback

Now that the community has something to work for, the members need a way to be able to collaborate, which would be the “mechanism of intercommunication.” The projects main public-facing website (cghmn.org) says that the best way for those wanting to run services on the network can get in touch with the creator on social media, or talk with the CGHMN team over IRC (Internet Relay Chat). That IRC is bridged to a Discord server and that Discord server is the primary way members (or team members) communicate (Figure 2).

Figure 2

<https://cghmn.org>



Note. The public-facing website cghmn.org is designed as a landing page for the Compu-Global-Hyper-Mega-Net. Includes information on how to get connected to the CGHMN, what services are running, what services are planned to be coming, communication information, etc. Linked on <https://cghmn.org> (figure 2) is the sign up page, which links to the Discord server and provides the most convenient way to get the information needed to connect you to the CGHMN.

Discourse communities use those intercommunication mechanisms to provide important information and solicit dialogue and communication between members. The intercommunication mechanism of Melzer's (2020) community, Meetup.com, was "how we recruited new members,

shared information about when and where we were playing, and communicated with each other outside of the night of the guitar jam” (p. 103). Of the two primary ways the CGHMN community can interact, Discord and IRC, Discord is the place to get important information and interact with other community members. On the Discord, you can view the status of the CGHMN and services running on it, new members can get help on connecting outside of the how-to article (https://wiki.cursedssilicon.net/wiki/How_to_Get_Connected), and there are numerous channels for members to talk about various topics. These mechanisms are not only how discourse communities interact, but it’s how members support their shared common goal.

The specialized languages and genres the community uses to further their goals

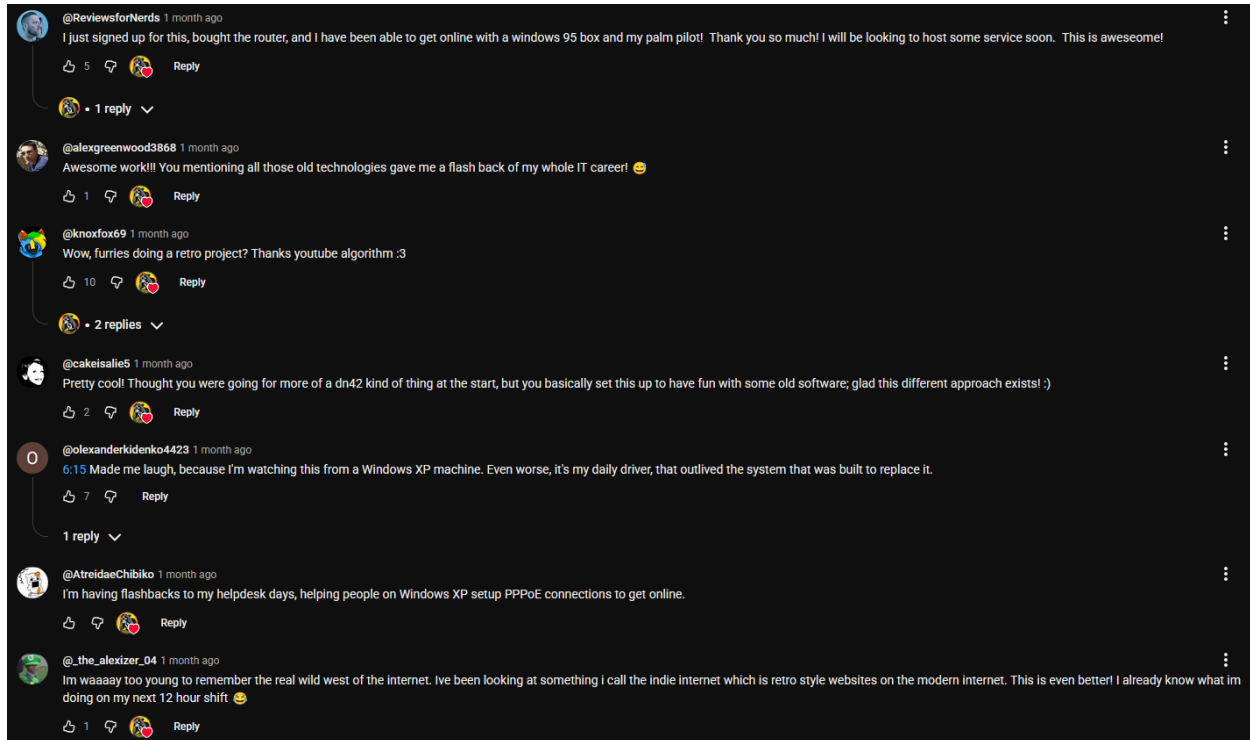
Outside of just communicating, discourse communities have various genres they use to support their common goal. Melzer (2020) brings up the idea of genre as a set of features, where music is organized into musical genres with other music because of the things they share like instruments and singing styles. But he expands upon genre to apply it to different styles of social action. “Rap also has social purposes, including resisting social oppression and telling the truth about social conditions that aren’t always reported on by news outlets. Like all genres, rap is not just a formula but a tool for social action” (p. 103). Genres, in the context of discourse communities, are the tools that the community uses to expand its ideas, wants, and goals.

To better understand the genres the CGHMN community uses, we also have to understand discourse communities’ specialized languages, or lexis. These specialized languages are used by discourse communities to convey specific and complex concepts to one another, something a random group of people would not be able to do as they do not have the necessary context from intercommunication. As brought up by Melzer (2020), they aren’t used to “show off or to try to discourage outsiders from joining our group” they were used to “make sure we

were all playing together effectively” (p.105). Or another example, a network cable technician must know terms like “TIA568B” and “8P8C” or “RJ-45” because the other technicians will use these terms with them to quickly convey what they need without having to explain how to put the cables together. (Though I wouldn’t qualify “cable technicians” as a discourse community, as every cable technician does not share the same common goal).

Now what are the specialized languages and genres used by the CGHMN community? There are several, but I’ll start with the IRC. Even though it’s also an intercommunication mechanism, it isn’t very effective as an intercommunication mechanism and is more of a novelty. But it does function well as a discourse community genre. It’s an outdated technology that is very easy to access once you actually get connected to the CGHMN, but can also be accessed from the real internet which makes it an easy goal for newcomers. To advertise the project, the CGHMN has been expo’d at a couple of computing conventions. I actually personally found out about the network from LinuxFest Northwest (LFNW) – which is a free-and-open-source / Linux convention put on and hosted by the InfoTech program once a year at Bellingham Technical College. There the creator had a talk to advertise the project and had an interactive demo of some games and the IRC in the sponsor / community expo area. To prove that the convention expositions are successful, just look at the YouTube recording of the LFNW talk (figure 3).

Figure 3

YouTube comments section

Note. Top 7 comments on the YouTube recording of Cursed Silicon's talk on the Compu-Global-Hyper-Mega-Net at LinuxFest Northwest.

These comments (figure 3) show that people are very excited about the idea. If you couldn't see the community in person and get hands-on, the CGHMN community can also effectively advertise itself on YouTube. But genres are more than just how the community advertises itself. Melzer (2020) discusses how his community learns songs together by using sheet music that features information so that all members of the community can play along. A more internal genre the CGHMN uses is retro game servers. These are super attractive to newcomers, as video games don't require that much effort to get people interested. To be able to play these games on the CGHMN, they need to get their system connected to the network. Actually connecting requires a lot of pre-existing knowledge and familiarity with computer networking terms (the lexis) for

setup to go smoothly. Some examples from the install guide (figure 3) are “TCP/IP (v4)”, “WireGuard” “lan”, “DHCP”, and “public key”.

Figure 4

https://wiki.cursed silicon.net/wiki/How_to_Get_Connected

Get connected - With OpenWRT

If you chose to go with an OpenWRT compatible router or want to run OpenWRT on typical x86 hardware/in a VM, you can follow these steps to get yourself connected to the CGHMN:

1. Update your OpenWRT install to the latest version to ensure all required packages are available and compatible.
2. Download [this script from GitHub](#) to your OpenWRT router: `wget https://cghmn.org/get-connected.sh`
3. Run the following commands on the router:
 1. `ash setup-cghmn.sh install-pkgs`
 1. If you run into issues at this step, please make sure you are on OpenWRT 25.12 or later - the setup script only supports version 25.12 and up.
 2. Reboot the router, this step is necessary if you intend to use the web UI for any other configuration or see the status of the CGHMN connection, to make LuCI recognize WireGuard connections.
 3. `ash setup-cghmn.sh init`
 4. You will be asked what network port you'd like to use for the Retro LAN. This is where you will plug in your retro machines to be part of the CGHMN. Choose a port that is not assigned to any OpenWRT interface like `lan` or `wan` or which not already part of a bridge and enter the Linux interface name, e.g. `eth1`, then press `[Enter]` to continue. If your router only has two ports and you're using one for WAN, then you first have to [enable the web UI and SSH access via the wan OpenWRT interface](#), remove the entire `lan` OpenWRT interface and the default `br-lan` or any other bridge the interface might belong to, to free the network port and continue the setup over the IP address your router got on its WAN side. If you only have a single Ethernet port, you're running on a router setup we can't really recommend, however you can configure VLANs and use a managed switch to both get a WAN DHCP address for internet access and have a separate VLAN for the Retro LAN bridge over a single port. This is commonly referred to as "router on a stick". Just enter the VLAN interface name here if you choose to go that route.
4. Now you will be given some information on the console, including a WireGuard public key. Head over to <https://signup.cghmn.org>, fill out the form there with your details like an e-mail address we can reach you under and copy over the WireGuard public key from the script output into the appropriate field on the web page. Once your request has been approved, you'll receive an e-mail with your CGHMN WireGuard connection details. Note: If you cannot copy-paste, for example, because you're on a VM VNC console, you can run `ash setup-cghmn.sh pubkey-qr` to get a QR code with your public key, which can be scanned with a phone, tablet or software QR code parser to get the key as copy-pastable text.
5. In the e-mail from your sign-up server post approval, you will receive a tunnel IPv4 address (`100.65.x.x/32`) and a routed IPv4 subnet (`100.68.x.0/24` and up). These will be needed on the next step
6. Run `ash setup-cghmn.sh set-tunnel-ip`, enter your new tunnel IP address with or without the `/32` subnet mask, enter your routed subnet **with** the CIDR netmask and supply your pre-shared key from the e-mail.
7. Once the script completed successfully, reboot the router to ensure all interfaces are up properly. After the reboot, your retro devices should receive an IP address in your routed IPv4 subnet on the Retro LAN port you chose above and be able to communicate with other machines on the CGHMN network.

Note. Screenshot of the “Get connected – With OpenWRT” section of the how-to-get connected page for the CGHMN, which is very technical and requires a lot of pre-existing knowledge to understand.

Little or no understanding of these terms does make getting connected a significant challenge, but at least the instructions are more or less designed to be copy-pasted just by looking for matching terms and following step-by-step. And again, there is a Discord server available for novices to get assistance. Once connected, there is an internal version of the public-facing webpage (figure 2) reachable with <http://cghmn.retro>. One minor genre on this page is a file share, which supports the community by making hard to find files very easy to access. There are plenty of games available, along with patches that hack multiplayer games (such as World of Warcraft) to be able to connect to the CGHMN version of their server. The community is supported through advertising and by having fun and interactive things to do once connected, but lots of specialized language and sometimes difficult access to needed information can limit the appeal of the community.

The relationship between expert and novice members

The final aspect of a discourse community is a presence of experts within the community. These are the people that assist novices to grow the community and make it last. There would be no CGHMN if we were all casual fans of old computers and just wanted to play video games as the very complex knowledge necessary to run the CGHMN requires lots of study and experience. Melzer's (2020) community supported their novice members by helping them with complex playing techniques and educating them on the group's social norms. Compared to his discourse community, the CGHMN is a bit different. Even though the expert members are able to provide assistance to novice members, the primary goal of expert members of the CGHMN community are not to make novice members experts, but rather to provide things to do on the CGHMN that everyone can play with. Nowhere in the community's methods of communication are detailed tutorials on running specific types of services like game and e-mail servers, no expert in the

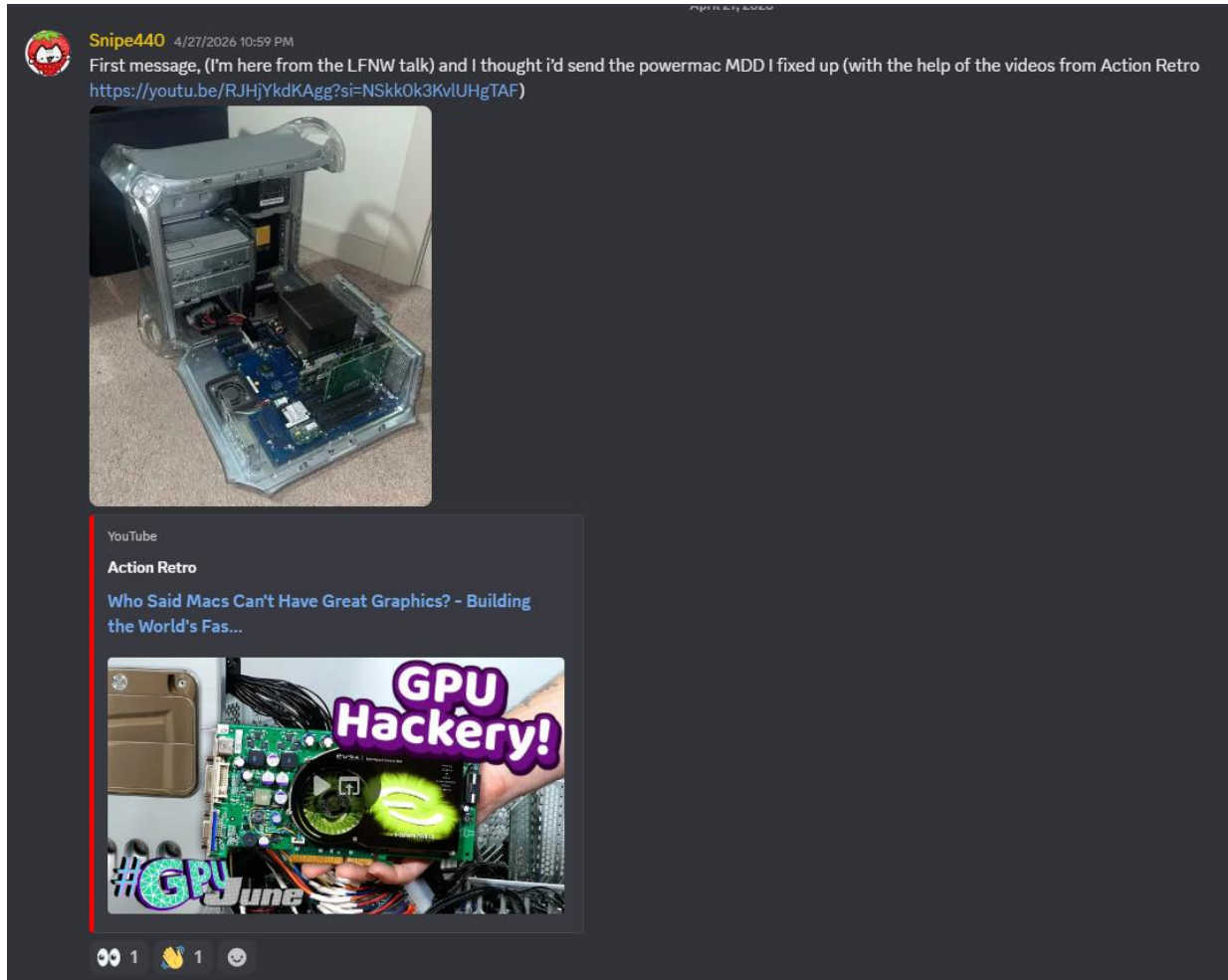
community expects to have to explain how to set them up either (though I'm sure they'd help if you asked). Fortunately most new members aren't novices when it comes to computer networking, as the primary audience for the community is retro computer enthusiasts where there is already some level of technical expertise. That's why the expert members are there to guide the novice members through the CGHMN, not teach them all the basics.

Becoming part of a discourse community

Now from my perspective on how I interact with the community. The first step to interacting with any discourse community is to use "mush fake discourse" before you are an expert in a community. In his article "Literacy, Discourse, and Linguistics: Introduction" Gee (1989) describes this type of interaction as "mush fake discourse" and defines it as "partial acquisition coupled with meta-knowledge and strategies to 'make do'" (p.13). Before you are actually a member of a discourse community, you need to get your foot in the door by using what knowledge you might already have of the community and a little bit of context to fake your way through initial interactions with that community. This step is important to learning how the community operates and to introduce you to that community. I mush faked by taking what I already know about the community and sent a picture of one of my projects (Figure 4) to learn what they were really interested in.

Figure 5

Screenshot of the “Cursed Silicon” Discord



This was how I got myself out there, If I really wanted to be a part of this community, I have to engage with it. To really become part of a discourse community, you must expand beyond your learned secondary discourse, and start using your primary discourse when interacting with the community. Gee (1989) describes primary discourse as the initial ways of speaking and interacting that we learn when we are children, when we are interacting with our first communities. These primary discourses are learned “not by overt instruction, but by being a member of a primary socializing group” (Gee, 1989, p.8). Think of how you interact with your

parents or sibling at home, they're going to be different than how you interact with strangers in public, or peers at school. Then there are the learned secondary discourses which Gee (1989) defines as the "various non-home-based social institutions-institutions in the public sphere, beyond the family and immediate kin and peer group" that we interact with and the discourses we learn (p.8). These secondary discourses are the ones we learn to be able to fit in when we're outside of our primary discourse environments, where we adopt their styles of communication to fit in. When joining a discourse community, first we must fake by using our secondary discourse community experience and existing knowledge of that discourse community to introduce ourselves. Then if that must faking is successful, we stop faking and fully adopt the discourse community's secondary discourse.

Conclusion

As defined by Swales, the Compu-Global-Hyper-Mega-Net operates well as a discourse community. Unlike a group of retro computing enthusiasts who just happen to play video games together, the CGHMN uses the aspects of a discourse community to effectively promote itself. I think discourse communities are successful when they allow people of a specific niche to find other like-minded people. The CGHMN achieves this through effective advertising in-person and online that encourages like-minded people to meet and interact with each other. I do notice a couple of limitations though. Instead of being a light barrier, the lexis of the CGHMN community can serve as a barrier if not already an expert in computer networking. Along with that, I think the websites where newcomers need to get information are a little messy and could be clearer. But if you are already familiar with computer networking and retro computing, CGHMN is a fun, engaging, and supportive discourse community.

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